



A cellular automata-based method for salt-and-pepper noise removal in images

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Abstract

Salt-and-Pepper Noise (SPN) is one of the major challenges in image restoration, as it tends to severely degrade structural details and textures, particularly under high noise conditions. In this paper, we propose a novel image denoising method based on Cellular Automata (CA). The proposed method leverages simple local rules and the inherent ability of CA to preserve spatial continuity, allowing for the iterative removal of SPN while effectively maintaining the structural characteristics of the original image. Experimental results on standard benchmark datasets show that the proposed method achieves higher Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index Measure (SSIM) compared to existing techniques. Furthermore, histogram and entropy analyses confirm that the intrinsic features of the images are well preserved after denoising. The proposed approach demonstrates stable performance even under severe noise conditions, without relying on complex learning models or high computational cost, making it highly suitable for real-time image restoration and practical applications such as embedded systems.

Keywords Image denoising · Image restoration · Salt-and-pepper noise · Cellular automata · Performance evaluation

List of abbreviations

CA	Cellular Automata
SPN	Salt-and-Pepper Noise
PSNR	Peak Signal-to-Noise Ratio
SSIM	Structural Similarity Index Measure
MF	Median Filter
ACWMF	Adaptive Center Weighted Median Filter

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DBA	Decision-Based Algorithm
NAFSMF	Noise Adaptive Fuzzy Switching Median Filter
NASEPF	Noise Adaptive Edge-Preserving Filter
INLM	Iterative Nonlocal Means
DAMF	Different Applied Median Filter
FSAP	Adaptive Type-2 Fuzzy Approach for SPN
DAF	Detail-Aware Filter

1 Introduction

Image denoising is an essential task in image processing and computer vision, aiming to restore image quality by eliminating noise while preserving key structural details. Noise can be introduced due to various factors, including sensor limitations, transmission errors, and environmental conditions, significantly degrading the visual quality of an image. The impact of noise is particularly critical in fields such as medical imaging, remote sensing, and digital photography, where high-quality images are crucial for accurate interpretation, diagnosis, and decision-making [15].

Among various types of noise, Salt-and-Pepper noise (SPN) is one of the most common and detrimental types, often arising due to errors in transmission, sensor faults, or environmental interference. SPN manifests as random occurrences of extreme pixel intensities (0 and 255), resulting in substantial information loss and making image restoration particularly challenging. As the noise density increases, the preservation of fine details and edges becomes significantly more difficult, necessitating robust denoising techniques [10].

A wide range of approaches has been proposed for the removal of SPN, spanning from traditional filtering techniques to advanced deep learning-based methods. In particular, median filter (MF)-based approaches [2, 6, 13] have been widely utilized due to their simplicity and computational efficiency. However, such methods tend to excessively smooth textures and edges, making it difficult to preserve fine structural details, especially under high noise densities.

To address these limitations, more advanced approaches, including decision-based algorithms (DBA) [19], fuzzy logic-based filters [20], and graph-based denoising techniques [25], have been explored. DBA selectively replaces noisy pixels using non-noisy neighbors, improving edge preservation but still struggling under extreme noise conditions. Fuzzy-based denoising techniques utilize fuzzy logic and probabilistic models to refine noise detection and removal, but they often require careful parameter tuning, limiting their adaptability. Graph-based denoising techniques model images as graphs, where pixels are nodes and edges represent relationships between pixels. These methods aim to preserve image structures by considering the global properties of the image but can be computationally intensive. More recently, deep learning-based denoising models such as Denoising Convolutional Neural Networks (DnCNN) [24] and Generative Adversarial Networks (GANs) [9] have achieved state-of-the-art performance by learning complex noise patterns from large-scale datasets. However, these approaches require substantial computational resources, limiting their applicability in resource-constrained environments. In response to these limitations, alternative lightweight SPN denoising methods have been proposed [3, 8], highlighting the growing demand for simple yet effective solutions.

In a different approach, Cellular Automata (CA) has emerged as a promising alternative for image processing tasks, including segmentation, edge detection, and denoising [1, 5, 11]. CA is a discrete computational model that evolves based on predefined local rules, making it particularly well-suited for iterative denoising processes. Unlike traditional filtering techniques that apply static transformations, CA-based approaches leverage spatial interactions and adaptive neighborhood processing to iteratively remove noise while maintaining image consistency.

In this paper, we introduce a novel CA-based image denoising method designed to effectively eliminate SPN while preserving essential image structures. By employing a structured neighbor selection mechanism, the proposed approach efficiently replaces noise pixels while maintaining spatial continuity. The proposed method improves noise removal performance through iterative filtering in multiple stages, and shows more stable and robust performance than existing filtering methods. Our work focuses on developing an efficient non-learning-based approach and evaluates its performance against methods within the same low-complexity category, rather than directly comparing with deep learning-based models that operate under fundamentally different computational assumptions.

Experimental results confirm that the proposed method outperforms traditional filtering techniques in terms of Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index Measure (SSIM), demonstrating competitive performance across various noise densities. The ability to balance computational efficiency and noise removal effectiveness highlights its potential as a reliable and practical solution for image restoration tasks.

2 Backgrounds and related works

2.1 Denoising methods for salt-and-pepper noise

Salt-and-Pepper noise (SPN) is a type of impulse noise characterized by random replacement of pixel values with either the minimum(0) or maximum(255) intensity, resulting in localized severe distortion. As the noise density increases, the original image information is rapidly degraded. SPN tends to be more pronounced in images captured under low-light conditions or when transmission errors occur during data transfer. Figure 1 illustrates an original image, an image corrupted with 50% SPN, and the corresponding denoised result.

Numerous methods have been proposed to remove SPN, including median filtering, statistical approaches, non-local means filtering, and machine learning-based techniques. One

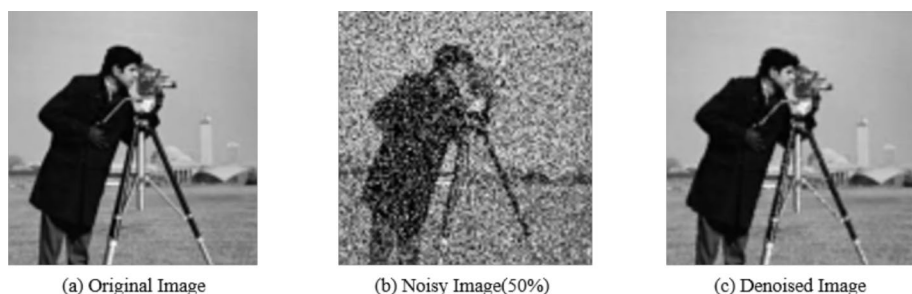


Fig. 1 An example of SPN denoising method

of the most widely used methods is the median filter(MF), which replaces noisy pixels with the median value of their surrounding neighborhood [2]. However, this method suffers from limitations under high noise density, as it often fails to preserve edge structures and fine image details. To address these weaknesses, the adaptive center-weighted median filter(ACWMF) was introduced [13], which dynamically adjusts the filtering window size for improved performance. Furthermore, several variants of median filtering, such as the Different Applied Median Filter(DAMF), aim to enhance denoising performance by modifying conventional median operations. Nonetheless, these methods may perform worse than traditional filters under specific conditions [6]. The Decision-Based Algorithm(DBA) replaces detected noisy pixels with either the mean or median of non-noisy neighboring values [19]. While offering simplicity and fast computation, this method is susceptible to performance degradation when the proportion of noisy pixels is high, as even the non-noisy pixels may be corrupted. To overcome this issue, the Noise Adaptive Fuzzy Switching Median Filter(NAFSMF) was developed [20]. By incorporating fuzzy logic, this method enables more accurate noise detection and pixel replacement. It decides whether to apply median filtering based on fuzzy rules, improving restoration performance. However, optimizing the fuzzy rule system requires additional computational overhead. The Noise Adaptive Edge-Preserving Filter(NASEPF) was designed to preserve edge structures during the denoising process [12]. While it enhances noise removal and maintains edge fidelity, its performance tends to decline in high-noise environments. Similarly, the Adaptive Type-2 Fuzzy Approach for Salt-and-Pepper Noise(FSAP) leverages a type-2 fuzzy system to achieve robust restoration [18]. This method offers better accuracy than traditional filters but incurs a significant computational cost due to the complexity of fuzzy optimization. Graph Signal Reconstruction-based filtering approaches formulate the denoising task as a graph signal processing problem, utilizing frequency characteristics to preserve edge information [25]. Although effective at maintaining image structures, these methods require high computational cost and an additional step for noise detection. On the other hand, the Iterative Nonlocal Means Filter(INLM) compares similar patches within the image to remove noise [22]. It excels at preserving textures but suffers from high computational complexity. The Detail-Aware Filter(DAF) is designed to suppress SPN while preserving important image details such as edges and textures [14]. However, it may be less effective in cases with specific noise patterns or structural irregularities.

In recent years, deep learning-based methods such as DnCNN and GANs have gained attention for SPN removal. DnCNN learns residual mappings through convolutional layers, while GANs generate restored images through adversarial training. Although these methods achieve impressive results, their dependency on large-scale training data and specialized hardware makes them unsuitable for lightweight deployment scenarios. Therefore, we exclude deep learning-based methods from our comparative analysis and focus on low-complexity algorithms that share similar computational constraints with the proposed method.

In this paper, we propose a novel denoising approach based on Cellular Automata (CA) that overcomes the limitations of existing methods. By leveraging the structural advantages of CA and its simple local rule-based computation, our method provides robust and efficient restoration performance without requiring any training process. In particular, the proposed method effectively preserves the structural characteristics of the original image even under high noise density conditions, where conventional methods typically suffer from significant performance degradation.

2.2 Cellular automata

Cellular Automata(CA) are discrete dynamic systems that generate complex patterns and system behaviors through the iterative application of simple local rules [23]. As a result of these properties, CA have been extensively applied in fields such as image processing, pattern recognition, cryptography, and biological modeling [4, 7, 16]. In CA, each cell determines the state of the next cell through the interaction with surrounding cells, including itself, which are called *neighborhood*. The interaction is called the *rule*.

Mathematically, a cellular automaton A is defined as:

$$A = (N, S, f)$$

where:

- N , the neighborhood is defined by $|N| = m$, where m is the number of neighbors for each cell i ; then $N = \{a_0, a_1, \dots, a_m\}$.
- S , it is the set of states that can take each of the cells.
- f , it is the local transition function of A of each cell defined by $f : S^m \rightarrow S$, that defines the evolution of A with respect to a neighborhood. Where $a_i^{t+1} = f(a_{i0}^t, \dots, a_{im}^t)$ that defines the next evolution of a cell with respect to its vicinity at the current time.

In two-dimensional CA, neighborhoods are typically categorized as either the Von Neumann neighborhood (Fig. 2(a)) or Moore neighborhood (Fig. 2(b)). In CA with Von Neumann neighborhood, the state of the center cell at time $t + 1$ is determined by the states of the current(time t) center cell and its four orthogonal neighbors(north, south, east, and west). In CA with Moore neighborhood, the influence extends to the four diagonal neighbors as well as forming a 3×3 grid centered around the center cell.

Figure 3 presents an example of a two-dimensional, two-state CA utilizing the Von Neumann neighborhood and binary states $s \in 0, 1$. The rule in this CA dictates that the next state of the center cell becomes 1 if exactly one or four of its neighbors are state 1; otherwise, the next state becomes 0.

This capacity to control pattern evolution through local interactions enables CA-based models to effectively balance simplicity and expressiveness, making them especially suitable for image denoising tasks. In such applications, CA can iteratively refine noisy pixels while preserving the structural integrity of image content.

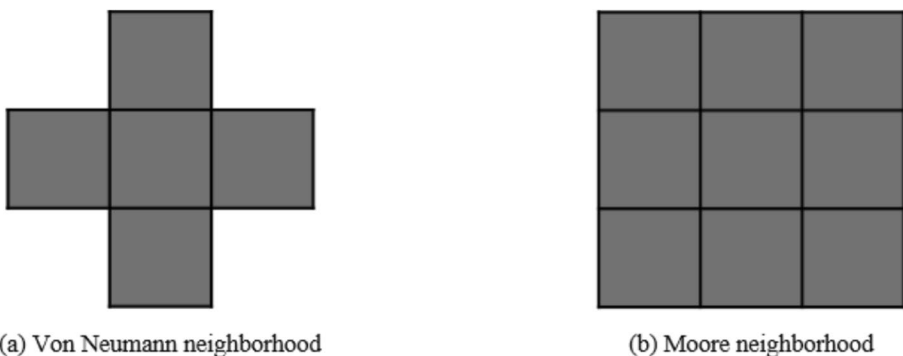


Fig. 2 Two types of 2D CA neighborhoods

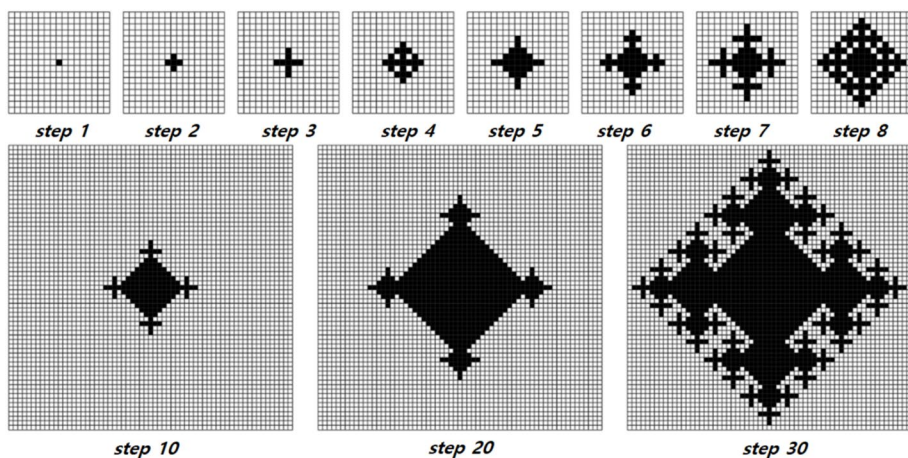


Fig. 3 An example of 2-dimensional Cellular automata [23]

In [17], a novel filtering method was proposed that combines fuzzy theory with cellular automata (CA) to effectively remove salt-and-pepper noise. Based on a 3×3 Moore neighborhood, their approach evaluates the statistical characteristics of neighboring pixels using Gaussian membership functions and applies a confidence-based weighted average, thereby removing noise while preserving edges and fine details. Their method benefits from the CA structure, which enables simple computations and parallel implementation, and demonstrated competitive PSNR and SSIM performance under high-density noise conditions (70–90%). In this paper, we proposed a model that outperforms their approach by employing a directional neighborhood selection strategy, which restores noisy pixels based on directionally coherent neighbors rather than relying on all surrounding pixels uniformly.

3 Proposed method

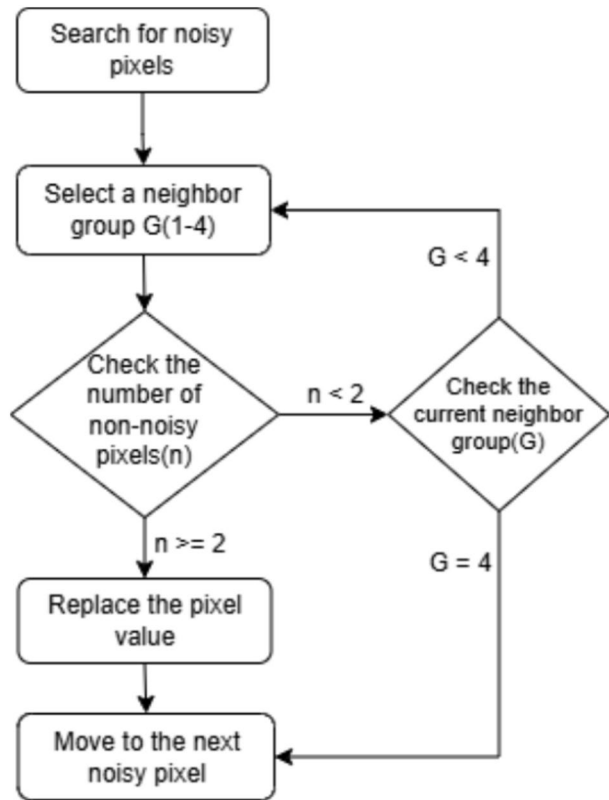
The proposed CA-based denoising algorithm operates in a sequential filtering process, as illustrated in Fig. 4. The pseudo-code is described as follows:

Algorithm 1 CA-based image denoising.

```

1: Input: Noisy image  $I$ 
2: Output: Denoised image  $I'$ 
3: Identify noisy pixels  $x_i$  in  $I$  such that  $x_i = 0$  or 255
4: repeat
5:   for each noisy pixel  $x_i$  do
6:     for each neighbor group  $G_k$  in order( $G_1 \rightarrow G_2 \rightarrow G_3 \rightarrow G_4$ ) do
7:       Count the number of non-noisy pixels( $n$ ) in  $G_k$ 
8:       if  $2 \leq n$  then
9:         Replace  $x_i$  with the mean of non-noisy pixels in  $G_k$ 
10:        break
11:      end if
12:    end for
13:  end for
14: until no noisy pixels remain

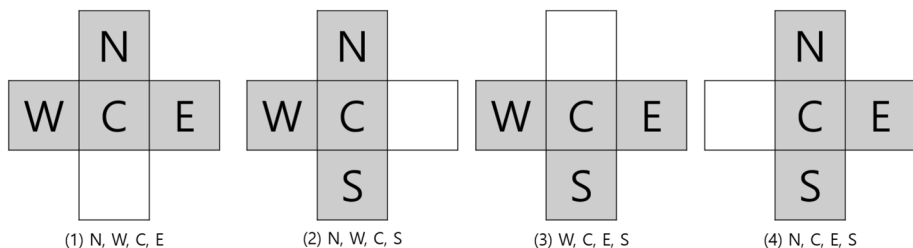
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Fig. 4 Flowchart of the proposed denoising method

The detailed steps for the above algorithm are as follows. The first step is to scan the entire image to identify noisy pixels x_i , defined as pixels with values of 0 or 255. These are considered to be corrupted by SPN.

Next, for each identified noisy pixel, a fixed sequence of four direction neighbor groups (denoted as G1 to G4) is applied, as illustrated in Fig. 5. These groups are defined based on the Von Neumann neighborhood structure. Applying groups in the predefined order ensures consistent filtering behavior across the image, reducing the variability that can arise from adaptive neighbor selection or texture-dependent neighbor selection.

This fixed-order approach plays a critical role in preserving spatial continuity while avoiding unnecessary complexity. By avoiding dynamically changing neighborhoods, the

**Fig. 5** Neighbor groups((1)-(4)) based on Von Neumann neighborhood

method maintains uniformity in processing and provides stable and predictable denoising performance—especially in high-noise regions or texture-rich areas. For each group G_k , the number of non-noisy (i.e., valid) pixels is counted. If the number of non-noisy pixels is at least two ($2 \leq n$), the noisy pixel x_i is replaced with the mean of the non-noisy values in that group. If not, the algorithm proceeds to the next neighbor group. This filtering mechanism can also be formulated mathematically using a CA transition function. Let C^t denote the pixel value of the center cell at time t (i.e., the noisy pixel x_i at time t). Similarly, let N^t, W^t, E^t, S^t represent the pixel values (ranging from 0 to 255) of the neighboring cells in the North, West, East, and South directions, respectively, at time t . Here, $u, v, x, y, z \in \{0, 1\}$ indicate whether each corresponding cell is non-noisy (1) or noisy (0). Using this notation, the update rule f_h , applied to C^t can be defined as follows:

$$\begin{aligned} C^{t+1} &= f_h(N^t, W^t, C^t, E^t, S^t) \\ &= \frac{u \cdot N^t + v \cdot W^t + x \cdot C^t + y \cdot E^t + z \cdot S^t}{\text{Hamming weight of } h} \end{aligned}$$

where $u, v, x, y, z \in \{0, 1\}$ indicate whether the corresponding neighboring cell is non-noisy cell with $h = u \cdot 2^4 + v \cdot 2^3 + x \cdot 2^2 + y \cdot 2^1 + z \cdot 2^0$.

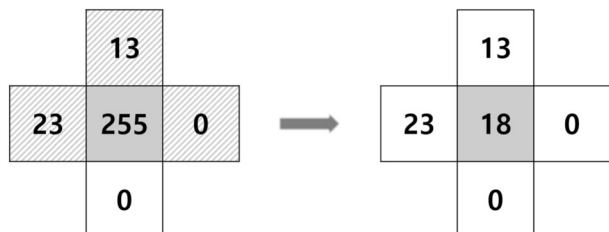
For example, in Fig. 6, the North and West neighbors are valid with $u = v = 1, x = y = z = 0$ (hence $h = 24$). Then the center pixel is updated as:

$$C^{t+1} = \frac{1 \cdot N^t + 1 \cdot W^t + 0 \cdot C^t + 0 \cdot E^t + 0 \cdot S^t}{\text{Hamming weight of } 24} = 18$$

If none of the groups provide a valid set of neighbors, the current noisy pixel remains unchanged in that iteration. The algorithm then continues to the next pixel. In order to preserve and restore the original information, the entire process is performed multiple times instead of necessarily removing all noise at once. This multi-pass strategy allows noise to be gradually removed, even in regions where valid information is sparse. It also helps in preserving structural information such as edges and textures.

Figure 7 shows the process of restoring pixel values of noisy pixels by the CA filter in a 5×5 pixel block with 11 noise pixels. As shown in Fig. 7, some noisy pixels are successfully replaced in the first iteration (e.g., the pixel at (1,1) is replaced by the average of 20 and 35, resulting in 28). However, pixels with insufficient valid neighbors (e.g., (1,2)) are left unchanged until a later iteration. After two iterations, all noise pixels in the example are successfully restored. This iterative and structured approach enables robust denoising performance, particularly in high-density noise conditions, while preserving important spatial features of the original image.

Fig. 6 The result of the CA filter with Group 1



14	20	35	0	33
35	255	0	255	255
75	86	0	0	45
52	47	85	77	0
84	0	109	255	255

(a)

→

14	20	35	34	33
35	28	0	255	39
75	86	86	61	45
52	47	85	77	61
84	80	109	98	255

(b)

→

14	20	35	34	33
35	28	32	37	39
75	86	86	61	45
52	47	85	77	61
84	80	109	98	80

(c)

Fig. 7 The result of denoising using proposed CA filter

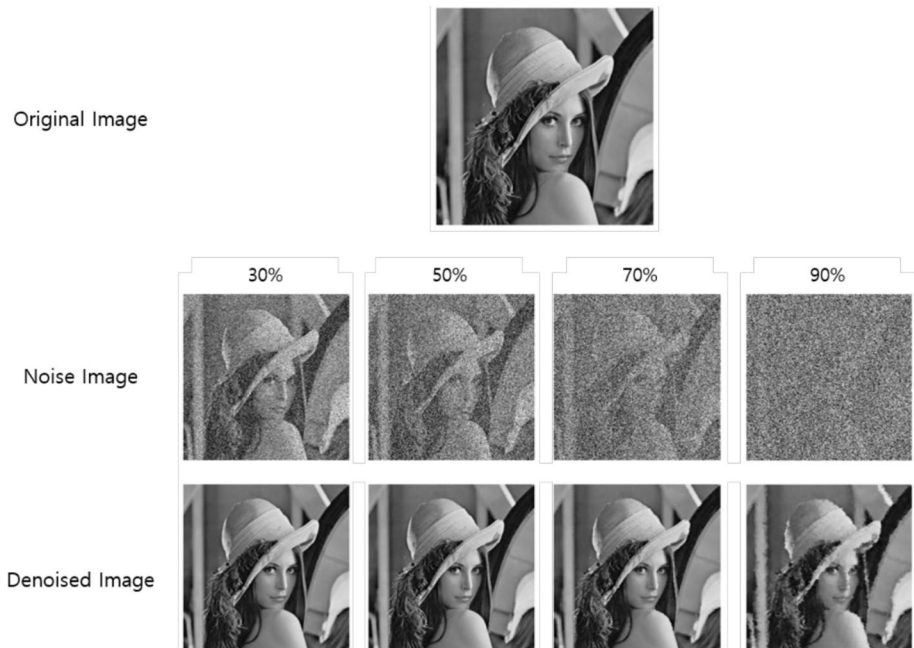


Fig. 8 The result of denoising using proposed CA filter

In addition, the computational complexity of our proposed method is $O(mn)$ for an $m \times n$ image, as it requires scanning the entire image and processing only noisy pixels with constant-time operations per pixel. This linear complexity, combined with the training-free nature of the algorithm, makes it suitable for real-time applications and resource-constrained environments.

Figure 8 shows the denoised results according to noise density using the proposed method.

4 Experimental results

This section presents a series of experiments and quantitative evaluations of the proposed CA-based denoising method.

We first applied varying levels of SPN to the Lena image and analyzed the corresponding entropy and histogram results to evaluate how well the proposed method restores the diversity and distribution of pixel values. This analysis demonstrates the proposed method's ability to recover meaningful image information.

Next, we tested the algorithm on 25 randomly selected images from the Berkeley Segmentation Dataset(BSD) as well as three standard test images—Lena, Baboon, Barbara—by adding SPN at different densities. Denoising performance was quantitatively evaluated using Peak Signal-to-Noise Ratio(PSNR) and Structural Similarity Index(SSIM), which measure the pixel-level accuracy and structural similarity with respect to the original images, respectively. Results showed that the proposed method consistently performs well across a wide range of image types and noise densities.

To further support its effectiveness, we compared the proposed method with several existing denoising techniques using both PSNR and SSIM metrics. These include traditional and advanced filters such as the Median Filter(MF), Adaptive Center Weighted Median Filter(ACWMF) [13], Noise Adaptive Fuzzy Switching Median Filter(NAFSMF) [20], an edge-preserving approach based on adaptive fuzzy switching median filter(NASEPF) [12], Iterative Nonlocal Means(INLM) [22], Different Applied Median Filter(DAMF) [6], Decision-Based Algorithm(DBA) [19], Adaptive Type-2 Fuzzy Filtering(FSAP) [18], and Detail-Aware Filter(DAF) [14].

4.1 Entropy and histogram analysis

In image denoising, maintaining a balance between noise removal and preservation of important image details(e.g., edges, textures) is critical. Entropy analysis helps evaluate how much information is retained or lost after denoising. Entropy, defined in information theory, measures the randomness or uncertainty in an image's pixel values and serves as an important indicator of image quality. The entropy H of a grayscale image is calculated as:

$$H = - \sum_{i=0}^{L-1} p_i \log_2 p_i$$

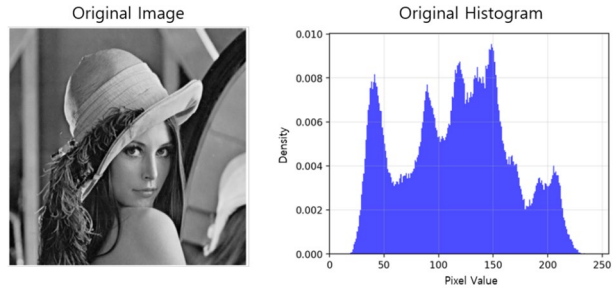
where p_i is the probability of a pixel having intensity i , and $L = 256$ for 8-bit grayscale images. A drastic drop in entropy after denoising suggests excessive information loss. Ideally, a good denoising method should result in high PSNR values and maintain entropy levels similar to the original image.

As shown in Table 1, the entropy of the noisy image is significantly different from the original, but the entropy of the denoised image closely approximates to the original. This indicates that information preservation is effectively achieved.

Table 1 Entropy comparison of original, noised, and denoised images

Noise density	Image	Entropy
50%	Original Image	7.4984
	Noisy image	5.2485
	Denoised Image	7.4862
90%	Noisy image	2.1184
	Denoised Image	7.4412

Fig. 9 The histogram of Lena image(gray-scale)



Histogram analysis provides a visual representation of pixel intensity distribution before and after denoising. In images corrupted by SPN, pixel values are highly concentrated at 0 and 255. Comparing histograms before and after denoising helps verify whether the noise has been successfully removed.

Figure 9 shows the histogram of the original Lena image(gray-scale), and Fig. 10 shows the histogram of the noisy and denoised versions of the original. The noisy image's histogram exhibits pronounced spikes at 0 and 255, whereas the denoised histogram closely resembles the original distribution, confirming that the proposed method effectively removes noise while preserving natural image characteristics.

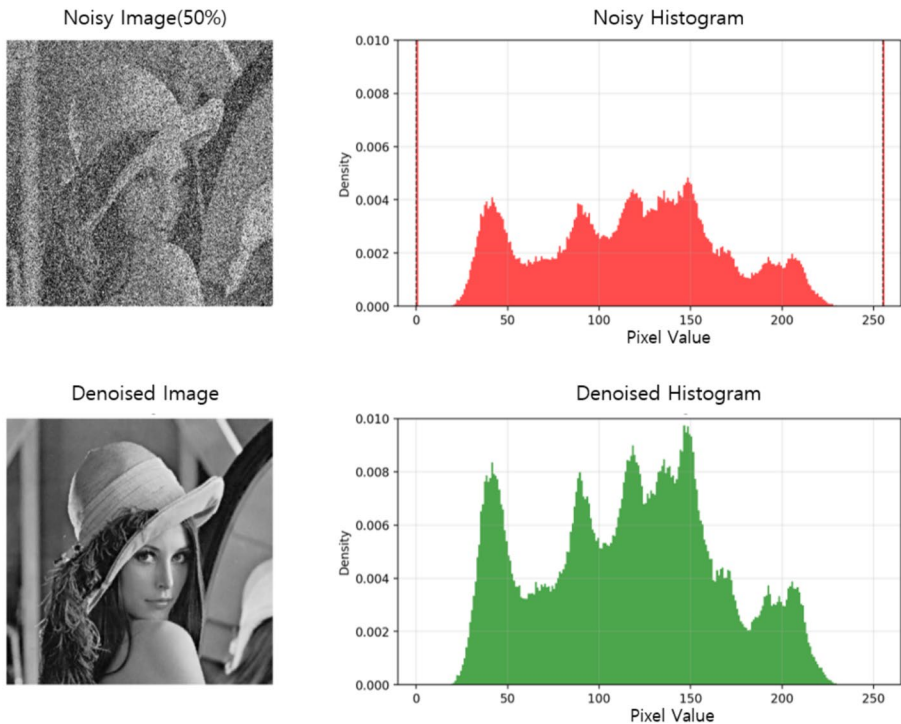


Fig. 10 The histogram for Noisy/Denoised images

4.2 Denoising performance on BSD and standard image dataset

PSNR is one of the most widely used metrics for evaluating image denoising performance. It quantifies the difference between the original and the restored image in terms of signal quality, based on the Mean Squared Error(MSE). PSNR is defined as:

$$\text{PSNR} = 10 \cdot \log_{10} \left(\frac{\text{MAX}^2}{\text{MSE}} \right)$$

where MAX denotes the maximum possible pixel value of the image(e.g., 255 for 8-bit grayscale images), and MSE represents the mean squared error between the original and the denoised images, calculated as:

$$\text{MSE} = \frac{1}{mn} \sum_{i=1}^m \sum_{j=1}^n [I(i, j) - K(i, j)]^2$$

Here, $I(i, j)$ refers to the pixel value of the original image and $K(i, j)$ refers to the corresponding pixel value in the denoised image. A higher PSNR value indicates better denoising performance, as it means the restored image is closer to the original. Typically, a PSNR value above 30 dB is considered to represent good image quality. A value above 40 dB suggests that the restored image is nearly indistinguishable from the original, while values below 20 dB indicate significant residual noise or degradation in the image.

SSIM(Structural Similarity Index) is another important metric for evaluating image quality after denoising, but unlike PSNR, it focuses on perceptual image quality based on the human visual system [21]. Instead of simply computing numerical differences between pixels, SSIM considers structural information such as luminance, contrast, and texture, which are crucial for human perception. SSIM is defined as:

$$\text{SSIM}(x, y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)}$$

In this equation, x represents the original image and y represents the denoised image. The terms μ_x and μ_y denote the mean intensities of the two images, while σ_x and σ_y are their variances. The term σ_{xy} represents the covariance between the two images. Constants C_1 and C_2 are included to stabilize the division and prevent instability when the denominators are close to zero. Unlike PSNR, SSIM does not measure pixel-wise error but instead captures structural similarities between images, making it a more reliable measure of visual quality. The SSIM value ranges from 0 to 1, where a value closer to 1 indicates higher structural similarity and, hence, better denoising performance.

Using a consistent set of 28 test images, noise was removed using the proposed method, and both PSNR and SSIM analyses were conducted to evaluate its performance. Table 2 presents the PSNR and SSIM values for the denoised images obtained from test images that were corrupted with varying levels of SPN. The results show that SSIM values vary depending on both the level of noise and the specific characteristics of each image. In general, as the noise density decreases, SSIM values tend to increase. For images with low noise

Table 2 SSIM and PSNR(dB) for denoising performance using proposed method

Image \ Noise	0%		30%		50%		70%		90%	
	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR
108073	0.99	36.12	0.95	30.83	0.90	27.79	0.81	25.24	0.61	21.80
118035	0.99	39.26	0.98	33.88	0.95	30.44	0.91	27.34	0.82	22.27
12074	0.99	40.82	0.98	35.12	0.95	31.80	0.91	28.56	0.79	22.76
135069	1.00	47.71	0.99	42.13	0.99	38.93	0.98	36.04	0.96	31.58
15088	0.99	38.50	0.96	32.50	0.92	28.88	0.83	25.66	0.62	22.05
176039	0.99	39.05	0.97	33.39	0.94	29.73	0.88	26.71	0.74	22.61
178054	0.99	41.30	0.96	35.56	0.92	32.44	0.85	29.70	0.67	26.00
189003	0.99	35.87	0.96	29.84	0.93	26.76	0.85	23.59	0.64	18.90
2092	0.99	39.65	0.96	34.41	0.91	31.43	0.85	29.11	0.74	26.31
22090	0.99	39.74	0.97	34.00	0.94	30.92	0.89	27.68	0.76	23.29
232038	0.98	37.11	0.94	31.86	0.88	29.06	0.80	26.75	0.65	23.77
24004	0.98	33.40	0.94	28.10	0.88	25.40	0.80	22.89	0.62	19.89
24063	0.99	42.85	0.98	37.57	0.97	34.22	0.94	31.16	0.86	26.44
26031	0.98	28.81	0.93	23.69	0.86	20.62	0.74	18.08	0.48	15.17
268002	0.99	38.69	0.97	33.27	0.94	30.05	0.88	27.43	0.73	23.42
299091	0.99	45.05	0.97	39.29	0.94	36.36	0.90	34.00	0.81	30.53
35010	0.99	37.25	0.96	31.67	0.92	28.21	0.85	25.13	0.64	20.18
365025	0.99	41.01	0.97	35.44	0.94	32.42	0.88	29.73	0.73	25.23
385028	0.98	34.79	0.91	29.82	0.83	26.96	0.71	24.72	0.50	21.76
61060	0.99	36.67	0.97	31.08	0.93	27.82	0.86	24.89	0.69	21.11
65010	0.99	36.51	0.95	30.97	0.90	27.91	0.83	25.39	0.64	21.98
65074	0.99	35.89	0.96	30.28	0.92	27.18	0.85	24.31	0.67	20.74
66075	0.99	41.23	0.97	35.88	0.95	32.97	0.91	30.45	0.83	27.11
94079	0.98	36.29	0.94	30.57	0.88	27.64	0.77	24.96	0.51	21.48
97017	0.99	36.29	0.96	31.08	0.93	28.27	0.86	25.41	0.73	21.70

density, SSIM values are consistently close to 1, indicating high structural similarity to the original image. However, as the noise density increases, the range of SSIM values becomes wider. This variation is attributed to differences in texture and size among the test images. A similar trend is observed with PSNR. For images with low noise density, the PSNR of the denoised results approaches 40 dB, indicating that the restored images are almost indistinguishable from the originals. Although PSNR gradually decreases as the noise density increases, most of the restored images still maintain acceptable visual quality, which proves the reliability of the proposed method.

The proposed model operates by replacing each noise pixel with a new value computed from the meaningful (non-noisy) pixels within its neighborhood. As the noise density increases, the number of valid pixels available in the local neighborhood decreases significantly, making it infeasible to restore all corrupted pixels in a single pass. Consequently, the algorithm employs an iterative approach, progressively stabilizing the neighboring pixels while gradually reducing the number of noise pixels until the entire image is fully restored.

Table 3 illustrates the reduction in residual noise ratio at each iteration for the Lena image corrupted with 70% noise density. The majority of corrupted pixels are restored during the early iterations, while the remaining iterations handle isolated noise clusters that require neighborhood stabilization.

Table 3 Noise pixel reduction per iteration at 70% noise density

Iteration	Noise Pixels	Residual Noise Ratio(%)
0 (Initial)	183,500	100.00
1	119,840	65.31
2	74,256	40.46
3	42,244	23.02
4	21,664	11.81
5	10,190	5.55
6	4,501	2.45
7	1,830	1.00
8	686	0.37
9	239	0.13
10	81	0.04
11	31	0.02
12	13	< 0.01
13	5	< 0.01
14	1	< 0.01
15	0	0.00

Table 4 Number of iterations required for complete noise removal at various noise densities

Noise (%)	10	20	30	40	50	60	70	80	90
Iterations	2	4	4	6	6	9	15	28	69

Table 4 presents the number of iterations required to fully restore the Lena image at various noise densities. At noise density of 70% and above, the required number of iterations increases sharply, as the probability of all neighboring cells being corrupted rises significantly, leaving fewer valid reference pixels available for restoration.

4.3 Performance comparison with existing denoising techniques

In this section, we proposed a novel CA-based method for removing SPN from images. To evaluate its effectiveness, we compared the proposed method with several conventional denoising techniques, including the Median Filter(MF), Adaptive Center Weighted Median Filter(ACWMF), Noise Adaptive Fuzzy Switching Median Filter(NAFSMF), Iterative Nonlocal Means(INLM), Different Applied Median Filter(DAMF), Decision-Based Algorithm(DBA), Adaptive Type-2 Fuzzy Approach for SPN(FSAP), and Detail-Aware Filter(DAF).

It is important to note that the comparative methods selected in this paper belong to the category of lightweight, non-learning-based algorithms. As discussed in Section 2.1, deep learning-based approaches operate under fundamentally different computational assumptions and are therefore excluded from this comparison. The proposed method operates without any training phase and relies solely on simple local rules, making it inherently lightweight and straightforward to implement without parameter tuning. The experimental results demonstrate that the proposed method achieves competitive performance compared to existing methods within this low-cost algorithm category.

Table 5 and Fig. 11, and Table 6 and Fig. 12, present the PSNR and SSIM values and their corresponding graphs, respectively, obtained from applying these denoising methods, including the proposed method, to four standard test images: Barbara, Lena, Baboon.

The proposed method consistently achieved higher PSNR and SSIM values compared to the existing approaches across all noise densities. In particular, it demonstrated stable and reliable performance even under high noise conditions. For images where edge preservation is critical—such as Lena—the method effectively maintained edge details, resulting in high SSIM scores. Similarly, for texture-rich images like Baboon and Barbara, the proposed method outperformed other techniques, showing strong capability in preserving fine image details. While many conventional models exhibited significant performance degradation as the noise density increased, the proposed method maintained reliable and consistent restoration performance even in highly noisy environments.

These results confirm that the proposed model achieves higher PSNR and SSIM values compared to existing methods, indicating that it effectively preserves both the original image content and its structural characteristics even after denoising. Notably, as the noise

Table 5 Denoising performance comparison(PSNR)

Image	Method	10%	20%	30%	40%	50%	60%	70%	80%	90%
Barbara	MF	24.13	22.74	21.88	21.17	20.28	19.18	16.19	11.88	7.69
	ACWMF	29.79	26.6	21.95	17.84	14.21	11.53	9.17	7.41	5.83
	DBA	33.56	30.69	28.68	27.07	25.45	23.72	21.89	19.54	16.5
	NAFSMF	34.97	31.78	29.78	28.21	27.1	25.77	24.73	23.44	20.82
	NASEPF	29.28	26.14	24.56	23.69	23.17	22.89	22.94	22.72	20.55
	INLM	35.41	31.47	28.7	27.48	26.23	25.67	25.51	24.86	23.07
	DAMF	34.97	31.78	29.78	28.21	27.12	25.8	24.82	23.64	21.79
	FSAP	35.04	31.64	29.35	27.26	25.4	23.23	21.2	18.87	16.49
	DAF	36.28	32.93	31.07	29.6	28.64	27.42	26.38	25.21	23.47
	OURS	37.31	34	32.16	30.62	29.44	28.2	27.07	25.54	23.27
Lena	MF	25.00	23.39	22.32	21.15	20.49	18.81	16.26	12.11	8.20
	ACWMF	33.45	28.29	22.53	18.26	14.79	11.94	9.68	7.76	6.35
	DBA	37.54	33.34	30.96	28.24	27.02	24.89	22.64	20.33	17.48
	NAFSMF	38.54	34.73	32.46	30.66	29.1	27.56	26.31	24.71	21.55
	NASEPF	27.99	24.84	23.4	22.47	22.13	22.17	22.72	23.03	21.25
	INLM	30.43	27.24	25.85	25.09	25.07	25.3	25.9	25.68	23.39
	DAMF	38.54	34.73	32.46	30.66	29.11	27.58	26.46	24.95	22.82
	FSAP	37.25	33.37	30.84	28.23	26.08	23.62	21.62	19.27	17.00
	DAF	39.25	35.73	33.46	31.63	30.3	28.76	27.77	26.36	24.15
	OURS	41.84	38.33	36.12	34.42	32.92	31.44	29.77	27.59	24.24
Baboon	MF	19.58	19.32	18.98	18.65	18.26	17.34	15.27	11.31	7.48
	ACWMF	25.76	23.45	20.16	16.81	13.75	11.14	9.02	7.24	5.80
	DBA	28.02	25.96	24.24	22.84	21.59	20.24	19.04	17.74	16.15
	NAFSMF	30.87	27.80	25.90	24.35	23.05	21.88	20.72	19.49	17.69
	NASEPF	26.56	23.62	22.03	21.01	20.36	19.94	19.57	19.05	17.55
	INLM	28.93	26.79	25.06	23.56	22.49	21.73	21.08	20.38	19.20
	DAMF	30.87	27.80	25.90	24.35	23.05	21.89	20.75	19.58	18.15
	FSAP	30.35	27.45	25.60	24.03	22.62	21.37	20.13	18.97	17.49
	DAF	31.23	28.04	26.08	24.53	23.33	22.35	21.41	20.49	19.35
	OURS	32.66	29.51	27.42	25.81	24.51	23.28	22.11	20.9	19.56

(unit : dB)

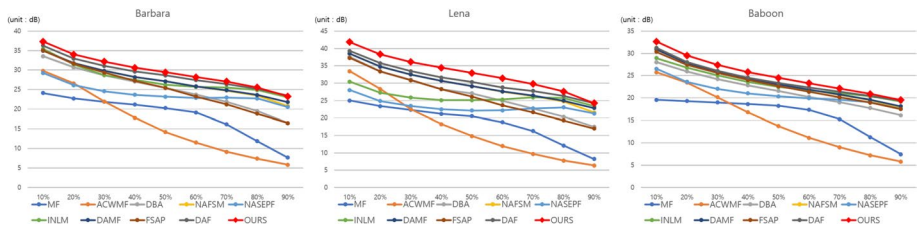


Fig. 11 Comparison with other models for denoising performance(PSNR)

Table 6 Denoising performance comparison(SSIM)

Image	Method	10%	20%	30%	40%	50%	60%	70%	80%	90%
Barbara	MF	0.7048	0.6969	0.6852	0.6749	0.6471	0.6084	0.4466	0.2035	0.0466
	ACWMF	0.9494	0.8918	0.7373	0.5042	0.2947	0.1513	0.0785	0.0377	0.0155
	DBA	0.9769	0.9507	0.9174	0.8797	0.8319	0.7729	0.6864	0.5678	0.3945
	NAFSMF	0.9790	0.9562	0.9288	0.8978	0.8659	0.8243	0.7781	0.7144	0.5882
	NASEPF	0.8877	0.8280	0.7883	0.7539	0.7224	0.6946	0.6665	0.6379	0.5625
	INLM	0.9701	0.9319	0.8837	0.8508	0.8184	0.7969	0.7794	0.7523	0.6743
	DAMF	0.9790	0.9562	0.9288	0.8978	0.8662	0.8251	0.7809	0.7207	0.6285
	FSAP	0.9798	0.9578	0.9265	0.8836	0.8289	0.7484	0.6433	0.5007	0.3509
	DAF	0.9834	0.9641	0.9427	0.9174	0.8949	0.8633	0.8242	0.7757	0.6898
	OURS	0.9849	0.9678	0.9491	0.9279	0.9039	0.8728	0.8328	0.7709	0.659
Lena	MF	0.7651	0.7565	0.7480	0.7332	0.7092	0.6514	0.4768	0.1922	0.0400
	ACWMF	0.9733	0.9172	0.7477	0.5051	0.2708	0.1411	0.0691	0.0365	0.0139
	DBA	0.9856	0.9653	0.9416	0.9076	0.8692	0.8118	0.7326	0.6272	0.4832
	NAFSMF	0.9872	0.9716	0.9540	0.9314	0.9070	0.8728	0.8301	0.7763	0.6457
	NASEPF	0.8310	0.7604	0.7202	0.6887	0.6689	0.6504	0.6437	0.6418	0.6035
	INLM	0.8871	0.8441	0.8183	0.7926	0.7817	0.7777	0.7852	0.7834	0.7247
	DAMF	0.9872	0.9716	0.9540	0.9314	0.9070	0.8734	0.8346	0.7848	0.6918
	FSAP	0.9857	0.9677	0.9428	0.9039	0.8501	0.7676	0.6655	0.5319	0.3946
	DAF	0.9887	0.9747	0.9588	0.9381	0.9176	0.8898	0.8596	0.8205	0.7461
	OURS	0.9892	0.9772	0.9637	0.9487	0.9309	0.9086	0.8786	0.8298	0.7294
Baboon	MF	0.3842	0.3813	0.3777	0.3726	0.3630	0.3336	0.2545	0.1092	0.0253
	ACWMF	0.9189	0.8578	0.7191	0.5211	0.3162	0.1706	0.0857	0.0412	0.0188
	DBA	0.9673	0.9287	0.8802	0.8234	0.7536	0.6681	0.5676	0.4500	0.3101
	NAFSMF	0.9704	0.9373	0.8989	0.8557	0.8031	0.7428	0.6688	0.5745	0.4240
	NASEPF	0.8633	0.7904	0.7342	0.6867	0.6388	0.5933	0.5462	0.4939	0.3964
	INLM	0.9141	0.8785	0.8306	0.7759	0.7224	0.6733	0.6225	0.5618	0.4605
	DAMF	0.9704	0.9373	0.8989	0.8557	0.8032	0.7432	0.6700	0.5789	0.4463
	FSAP	0.9683	0.9345	0.8929	0.8407	0.7700	0.6868	0.5809	0.4632	0.3315
	DAF	0.9711	0.9367	0.8956	0.8490	0.7949	0.7362	0.6666	0.5857	0.4724
	OURS	0.9772	0.9506	0.9186	0.8817	0.8374	0.7814	0.7086	0.6083	0.4568

density increases, the proposed method continues to outperform conventional techniques, maintaining higher PSNR and SSIM scores. This highlights the model's strong potential and practical applicability in high-noise environments.

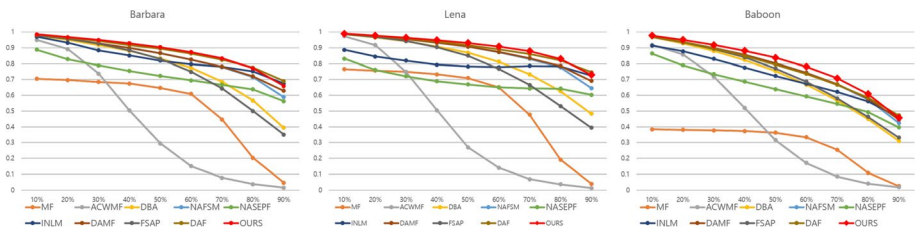


Fig. 12 Comparison with other models for denoising performance(SSIM)

5 Conclusion

In this paper, we proposed a novel image denoising method for removing Salt-and-Pepper noise by leveraging the simple rule structure and spatial continuity preservation characteristics of CA. The proposed approach applies a fixed sequence of neighborhood groups to effectively eliminate noise while maintaining the structural integrity of the original image. Experimental results on various standard images and datasets demonstrated that the proposed method achieved higher PSNR and SSIM values compared to traditional methods within the low-cost, non-learning-based algorithm category. In particular, the method showed consistent and reliable performance even under high noise conditions. Unlike deep learning-based approaches that demand significant computational resources and extensive training data, the proposed method provides competitive performance with minimal computational overhead, making it highly suitable for real-time processing and embedded system applications. For future work, we aim to enhance performance by incorporating adaptive neighborhood structures and optimizing computational efficiency. Additionally, we plan to explore the method's applicability to other types of noise beyond Salt-and-Pepper noise.

Author Contributions Gil-Tak Kong developed the Cellular Automata model and implemented the simulations. Un-Sook Choi supervised the project, interpreted the results, and contributed to the writing and editing of the manuscript.

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Availability of data and materials The datasets used in this study, including the BSD dataset and standard benchmark images such as Lena and Barbara, are publicly available and can be accessed through standard image repositories.

Declarations

Ethics approval and consent to participate Not applicable.

Consent for publication Not applicable.

Competing interests The authors declare that they have no competing interests.

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